

Algorithm 1 Curriculum Model Averaging (CMA)

1: **Input:** Unsorted dataset D , quality scoring function $Q(\cdot)$, training steps T , warmup steps T_w , peak learning rate η_{peak} , number of checkpoints to average k , averaging decay hyperparameter α , checkpointing interval s .

2: **Output:** Final model parameters $\bar{\theta}_{final}$.

3: # Stage 1: Data Scheduling

4: Sort dataset D to create D_{sorted} where for any samples x_i, x_j , if $i < j$, then $Q(x_i) \leq Q(x_j)$.

5: # Stage 2: Warmup-Constant LR Training

6: Initialize model parameters θ_0 .

7: **for** $t = 1$ **to** T **do**

8: **if** $t \leq T_w$ **then** ▷ Linear warmup

9: $\eta_t \leftarrow \eta_{peak} \cdot (t/T_w)$

10: **else** ▷ Constant LR

11: $\eta_t \leftarrow \eta_{peak}$

12: **end if**

13: Fetch next data batch B_t from D_{sorted} .

14: $\theta_t \leftarrow \text{OptimizerUpdate}(\theta_{t-1}, \eta_t, B_t)$ ▷ e.g., Adam

15: **if** $t \in \{T - (k-1)s, \dots, T - s, T\}$ **then**

16: Save checkpoint θ_t .

17: **end if**

18: **end for**

19: # Stage 3: Model Averaging, e.g., EMA or SMA.

20: Let the set of saved checkpoints be $\{\theta_{T-is}\}_{i=0}^{k-1}$.

21: EMA: $\bar{\theta}_{final} \leftarrow \frac{\sum_{i=0}^{k-1} \alpha^i \theta_{T-is}}{\sum_{i=0}^{k-1} \alpha^i}$ ▷ SMA: $\bar{\theta}_{final} \leftarrow \frac{\sum_{i=0}^{k-1} \theta_{T-is}}{k}$

22: **return** $\bar{\theta}_{final}$

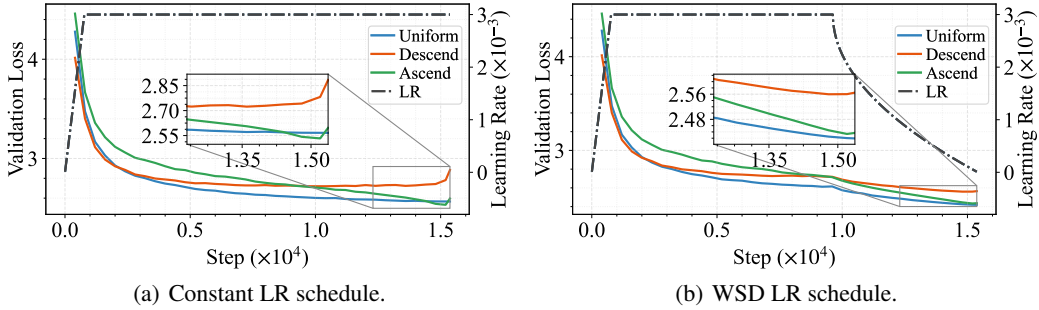


Figure 8: The benefits of a data curriculum using PreSelect scores also diminish. We show the validation loss curves for constant and WSD LR schedules under different data schedules, including uniform, ascending, and descending orders by PreSelect scores. Overall, the ascending curriculum outperforms the uniform baseline under a constant schedule, but cannot match it under the WSD LR schedule. The final validation loss of the data curriculum is higher than that of the uniform-ordering baseline, likely because the score metrics are not perfectly targeted to the validation set.

B DISCUSSION

B.1 COMPARISON WITH RELATED WORK

In this section, we situate our findings in the context of prior research on curriculum learning and discuss our interpretation compared with previous work on the loss landscape. We discuss how our experimental setup differs from related approaches and how our findings elucidate the interplay between learning rate (LR) scheduling, data ordering, and data quality.

Table 6: Downstream performance for experiments with PreSelect score ascending data. Our proposed methods (using WA) are highlighted in gray. **WA**: Weight Averaging (EMA: Exponential, SMA: Simple). **LRS**: Learning Rate Schedule (WSD: WSD with decay to 1×10^{-5} , Const: Constant LR, WSMD: WSD with moderate decay to 1×10^{-3}). **Core**: Average score on the first four, high signal-to-noise tasks (MMLU, ARC-c, ARC-e, CSQA). Both Core and Avg. scores are annotated with a subscript indicating the performance change relative to the baseline (first row). Subscripts in **bold green** indicate an improvement of ≥ 0.5 , **light green** an improvement of > 0 , and **red** a decrease.

WA	Order	LRS	MMLU	ARC-c	ARC-e	CSQA	Core	OBQA	PIQA	SIQA	Wino.	Avg.
X	Ascend	WSD	31.12	35.79	57.89	48.81	43.40	41.00	71.82	46.21	58.41	48.88
EMA	Ascend	Const	31.85	37.46	61.05	49.39	44.94 _{+1.54}	38.40	70.51	45.34	55.33	48.67 _{-0.21}
EMA	Ascend	WSMD	31.98	39.80	61.93	49.39	45.77 _{+2.37}	39.60	70.78	45.60	55.96	49.38 _{+0.50}
SMA	Ascend	WSMD	31.99	39.46	62.11	50.04	45.90 _{+2.50}	39.40	71.06	46.06	55.96	49.51 _{+0.63}

Table 7: Downstream performance for experiments on WebOrganizer dataset (Wettig et al., 2025). Our proposed methods (using WA) are highlighted in gray. **WA**: Weight Averaging (EMA: Exponential, SMA: Simple). **LRS**: Learning Rate Schedule (WSD: WSD with decay to 1×10^{-5} , Const: Constant LR, WSMD: WSD with moderate decay to 1×10^{-3}). **Core**: Average score on the first four, high signal-to-noise tasks (MMLU, ARC-c, ARC-e, CSQA). Both Core and Avg. scores are annotated with a subscript indicating the performance change relative to the baseline (first row). Performance changes are color-coded: **bold green** (≥ 0.5 improvement), **light green** (> 0 improvement), and **red** (decrease).

WA	Order	LRS	MMLU	ARC-c	ARC-e	CSQA	Core	OBQA	PIQA	SIQA	Wino.	Avg.
X	Uniform	WSD	28.92	34.45	47.72	47.83	39.73	36.60	72.03	43.76	56.67	46.00
	Ascend	WSD	29.09	32.78	51.58	47.42	40.22 _{+0.49}	38.00	72.14	44.73	55.09	46.35 _{+0.35}
EMA	Uniform	WSMD	28.28	34.11	47.89	48.81	39.77 _{+0.04}	39.20	71.65	43.76	55.56	46.16 _{+0.16}
EMA	Ascend	WSMD	28.56	31.10	50.88	48.89	39.86 _{+0.13}	39.60	71.44	44.11	56.75	46.42 _{+0.42}
EMA	Uniform	Const	28.03	33.44	47.72	47.42	39.15 _{-0.58}	40.60	70.78	43.65	55.72	45.92 _{-0.08}
EMA	Ascend	Const	29.32	33.11	55.09	48.89	41.60 _{+1.87}	38.40	71.00	45.29	55.41	47.06 _{+1.06}

Comparison with Prior Curriculum Learning Studies. Previous work on curriculum learning for large-scale language model (LLM) pretraining has often overlooked the interaction between data ordering and the learning rate schedule, typically adopting cosine LR schedules with low peak values on the order of 10^{-4} (Wettig et al., 2024; Dai et al., 2025; Zhang et al., 2025). For example, Wettig et al. (2024) reported a modest 0.6% improvement with a low-to-high quality curriculum but also found a counterintuitive 0.5% improvement with a reverse (descending-order) curriculum. Two factors may explain this paradox. First, in a low-peak-LR regime, LR decay can prematurely reduce the effective learning rate, narrowing the performance gap between curriculum and reverse-curriculum orders. Second, the data-quality metric itself may lack self-consistency, yielding noisy rankings.

In contrast, our experiments demonstrate a consistent performance drop for the reverse curriculum—especially under a constant LR schedule—and a smaller but still negative effect under schedules with LR decay (Table 8). These results indicate that our data-quality metric produces more self-consistent orderings and that LR decay indeed diminishes the benefits of a curriculum. While a direct quantitative comparison is not possible due to differences in scale and benchmarks, our best configuration (*Const+SMA*) achieves a relative improvement of over 2.7% compared to a comparable baseline (*Cos+Uniform*), demonstrating a substantially stronger effect than previously reported.

Limitations of the Data Folding Strategy. To address the limited gains of vanilla data curricula, recent work introduced the *folding* strategy, which divides the dataset into several consecutive folds and applies sorting within each stage (Dai et al., 2025; Zhang et al., 2025). Following Dai et al. (2025), we tested the three-fold configuration they identified as optimal. These experiments are conducted with a 0.5B model. We replicate their observation that folding improves performance under a low peak LR (1×10^{-4}) but find that this benefit diminishes—and even reverses—under a higher peak LR (3×10^{-3}), as shown in Table 9. The higher-LR setting shows better overall performance than the low-LR setting in our experiments. These findings, as well as results of 1.5B models in Section 4, suggest that folding is not robust across different model scales or learning-

Table 8: Performance comparison across learning rate schedules (Constant, Cosine) and data orders (Uniform, Ascend, Descend) on various downstream benchmarks. Results show that descending-order curricula perform worse than uniform ordering, and the performance gap narrows when LR decay is applied.

LR Schedule	Order	MMLU	ARC-c	ARC-e	CSQA	OBQA	PIQA	SIQA	Wino.	Avg.
Constant	Uniform	30.30	30.43	55.61	49.80	44.60	70.29	45.19	56.20	47.80
	Ascend	31.14	39.46	61.23	49.96	43.00	70.51	43.14	56.51	49.37
	Descend	29.43	33.11	45.96	45.21	41.00	69.86	44.98	56.75	45.79
Cosine	Uniform	30.49	38.13	59.47	49.14	42.20	71.87	45.19	56.51	49.13
	Ascend	30.80	39.80	59.12	51.27	42.60	71.55	45.65	57.06	49.73
	Descend	29.51	34.11	52.98	48.81	42.60	72.42	45.45	55.17	47.63

Table 9: Effect of the *folding* strategy under different peak learning rates. The benefit observed at a low LR (1×10^{-4}) diminishes or reverses at a higher LR (3×10^{-3}), indicating the limited robustness of folding across scales.

Order	Strategy	Peak LR	MMLU	ARC-c	ARC-e	CSQA	OBQA	PIQA	SIQA	Wino.	Avg.
Uniform	–	1×10^{-4}	25.70	28.43	37.72	34.32	30.20	61.81	40.99	50.83	38.75
Ascend	Sorting	1×10^{-4}	26.57	28.76	38.42	35.63	28.80	61.97	41.40	50.36	38.99
Ascend	Folding	1×10^{-4}	25.69	29.43	38.77	35.22	32.20	61.43	40.99	50.04	39.22
Uniform	–	3×10^{-3}	28.68	33.78	50.35	45.95	36.60	68.66	43.65	53.35	45.13
Ascend	Sorting	3×10^{-3}	27.78	37.12	47.89	44.47	37.40	67.85	43.30	55.80	45.20
Ascend	Folding	3×10^{-3}	28.33	33.11	48.25	43.82	38.80	69.21	43.76	52.88	44.77

rate settings. Its apparent gains may reflect compensation for a suboptimal LR schedule rather than a fundamental advantage. The limitations of the folding strategy and the strengths of our design underscore the importance of jointly considering curriculum design and LR scheduling.

Interpretation via the Loss Landscape. Our interpretation aligns with the river-valley model of the loss landscape (Wen et al., 2025), which characterizes optimization as progress along two primary directions: the *signal* (river) direction, where loss decreases gradually, and the *noise* (valley) direction, where loss oscillates sharply. We extend this framework by positing that data quality influences the gradient’s components: high-quality data are assumed to provide a stronger, more stable signal direction and less noise, whereas low-quality data provide a weaker signal and induce more noise. In the context of a data curriculum, as data quality increases, the update direction becomes more dominated by the stable signal component. This mechanism facilitates faster convergence, as observed in our experiments (Figures 1(a) and 8(a)). In the river-valley model, when LR decay is applied, the optimizer settles toward the valley floor, reducing noise but also slowing progress along the signal direction, thus underusing the signal from high-quality data.